CROSS-CHAIN STATE VERIFICATION METHOD, APPARATUS, ELECTRONIC DEVICE, AND STORAGE MEDIUM FOR BLOCKCHAIN

TECHNICAL FIELD

The present invention relates to the technical field of blockchain technology, and in particular, to a cross-chain state verification method, apparatus, electronic device, and storage medium for blockchain.

BACKGROUND ART

A blockchain is a decentralized network composed of distributed multiple nodes. Blockchain interoperability is a core infrastructure challenge currently facing the Web3 ecosystem. With the parallel development of heterogeneous blockchains (Ethereum, Bitcoin, Solana, Polygon, Arbitrum, etc.), the demand for cross-chain communication is increasing. Existing cross-chain communication for blockchains is implemented through centralized cross-chain bridges, oracle networks, light client verification, and other methods. These methods have issues such as collusion or other risks in cross-chain state verification, resulting in low security.

It is an object of the present invention to substantially overcome or ameliorate one or more of the above-mentioned drawbacks, or at least provide a useful alternative.

SUMMARY OF THE INVENTION

Embodiments of the present invention provide a cross-chain state verification method, apparatus, electronic device, and storage medium for blockchain, to address the problem of low security in existing cross-chain state verification processes.

According to a first aspect, the present invention provides a cross-chain state verification method for blockchain that comprises:
obtaining a state assertion submitted by a user on a source chain, wherein the state assertion is an assertion regarding verifiable data on the source chain;
generating a verifiable credential, wherein the verifiable credential comprises a credential identifier, a decentralized identifier, the state assertion, and a content hash, wherein the content hash is a result of performing a hash operation on the state assertion;

digitally signing the verifiable credential based on a user private key corresponding to the decentralized identifier to obtain a digitally signed verifiable credential; and
storing the digitally signed verifiable credential in a distributed storage system, such that a user on a destination chain can perform state verification on the verifiable data based on the digitally signed verifiable credential.

According to a second aspect, the present invention provides a cross-chain state verification method for blockchain that comprises:
obtaining a cross-chain business request submitted by a user on a destination chain, wherein the cross-chain business request comprises a business identifier and a credential identifier;
transmitting a verification request carrying the credential identifier to validators, such that the validators access a distributed storage system based on the credential identifier, determine a digitally signed verifiable credential, perform verification based on the verifiable credential, determine verification results, and receiving the verification results returned by the validators;
executing a business operation corresponding to the business identifier when the verification results returned by the validators satisfy an executable condition;
wherein the verifiable credential comprises the credential identifier, a decentralized identifier, a state assertion, and a content hash; the content hash is a result of performing a hash operation on the state assertion; the digital signature is a result of digitally signing the content hash based on a user private key corresponding to the decentralized identifier; and the state assertion is an assertion regarding verifiable data on a source chain.

According to a third aspect, the present invention provides a cross-chain state verification apparatus for blockchain that comprises:
an assertion obtaining module, configured to obtain a state assertion submitted by a user on a source chain, wherein the state assertion is an assertion regarding verifiable data on the source chain;
a credential generation module, configured to generate a verifiable credential, wherein the verifiable credential comprises a credential identifier, a decentralized identifier, the state assertion, and a content hash, wherein the content hash is a result of performing a hash operation on the state assertion;
a digital signature module, configured to digitally sign the verifiable credential based on a user private key corresponding to

the decentralized identifier to obtain a digitally signed verifiable credential; and
a credential storage module, configured to store the digitally signed verifiable credential in a distributed storage system, such that a user on a destination chain can perform state verification on the verifiable data based on the digitally signed verifiable credential.

According to a fourth aspect, the present invention provides a cross-chain state verification apparatus for blockchain that comprises:
a request obtaining module, configured to obtain a cross-chain business request submitted by a user on a destination chain, wherein the cross-chain business request comprises a business identifier and a credential identifier;
a credential verification module, configured to transmit a verification request carrying the credential identifier to validators, such that the validators access a distributed storage system based on the credential identifier, determine a digitally signed verifiable credential, perform verification based on the verifiable credential, determine verification results, and receive the verification results returned by the validators;
an operation execution module, configured to execute a business operation corresponding to the business identifier when the verification results returned by the validators satisfy an executable condition;
wherein the verifiable credential comprises the credential identifier, a decentralized identifier, a state assertion, and a content hash; the content hash is a result of performing a hash operation on the state assertion; the digital signature is a result of digitally signing the content hash based on a user private key corresponding to the decentralized identifier; and the state assertion is an assertion regarding verifiable data on a source chain.

According to a fifth aspect, the present invention provides an electronic device that comprises a memory, a processor, and a computer program stored in the memory and executable on the processor, wherein when the processor executes the computer program, the cross-chain state verification method for blockchain described above is implemented.

According to a sixth aspect, the present invention provides a computer-readable storage medium that stores a computer program, wherein when the computer program is executed by a processor, the

cross-chain state verification method for blockchain described above is implemented.

Embodiments of the present invention provide a cross-chain state verification method, apparatus, electronic device, and storage medium for blockchain. Based on verification using verifiable credentials, the invention provides the following three anti-fraud mechanisms to avoid collusion or other risks and ensure verification security: First, since all verifiable credentials on IPFS possess immutable content hashes, validators must verify the consistency between the content hash of the verifiable credential stored in IPFS and its state assertion content before verification. Any modification to the verifiable credential will cause the content hash in IPFS to change,

CLAIMS:

1. A cross-chain state verification method for blockchain, comprising:
obtaining a state assertion submitted by a user on a source chain, wherein the state assertion is an assertion regarding verifiable data on the source chain;
generating a verifiable credential, wherein the verifiable credential comprises a credential identifier, a decentralized identifier, the state assertion, and a content hash, wherein the content hash is a result of performing a hash operation on the state assertion;
digitally signing the verifiable credential based on a user private key corresponding to the decentralized identifier to obtain a digitally signed verifiable credential; and
storing the digitally signed verifiable credential in a distributed storage system, such that a user on a destination chain can perform state verification on the verifiable data based on the digitally signed verifiable credential.

2. A cross-chain state verification method for blockchain, comprising:
obtaining a cross-chain business request submitted by a user on a destination chain, wherein the cross-chain business request comprises a business identifier and a credential identifier;
transmitting a verification request carrying the credential identifier to validators, such that the validators access a distributed storage system based on the credential identifier, determine a digitally signed verifiable credential, perform verification based on the verifiable credential, determine verification results, and receiving the verification results returned by the validators;
executing a business operation corresponding to the business identifier when the verification results returned by the validators satisfy an executable condition;
wherein the verifiable credential comprises the credential identifier, a decentralized identifier, a state assertion, and a content hash; the content hash is a result of performing a hash operation on the state assertion; the digital signature is a result of digitally signing the content hash based on a user private key corresponding to the decentralized identifier; and the state assertion is an assertion regarding verifiable data on a source chain.

3. The method according to claim 2, wherein, transmitting the verification request carrying the credential identifier to validators comprises:
determining a required number of validators based on a business type corresponding to the business identifier, and transmitting the verification request carrying the credential identifier to validators matching the required number of validators.

4. The method according to claim 2, wherein, performing verification based on the digitally signed verifiable credential and determining verification results comprises:
performing content integrity verification based on the state assertion and the content hash in the verifiable credential to obtain an integrity verification result;
performing signature validity verification using a user public key corresponding to the decentralized identifier in the verifiable credential to obtain a signature verification result;
performing state consistency verification on the state assertion in the verifiable credential and the verifiable data to obtain a state verification result; and
determining the verification result based on the integrity verification result, the signature verification result, and the state verification result.

5. The method according to claim 4, wherein, after obtaining the state verification result, the method further comprises:
when the state verification result indicates verification failure, determining the state assertion as a fraudulent assertion and confiscating collateral assets corresponding to the user who submitted the state assertion.

6. The method according to claim 2, wherein, the executable condition comprises: a number of voting validators being greater than a validator quantity threshold corresponding to the business identifier, and a plurality of verification results satisfying a verification consensus condition;
wherein the number of voting validators is the number of validators that actually participated in verifying the verifiable credential; and the verification consensus condition is a pre-configured condition for evaluating whether a plurality of verification results have reached consensus approval.

7. The method according to claim 6, wherein, the verification consensus condition comprises: a consensus ratio being greater than a consensus ratio threshold corresponding to the business identifier; wherein the consensus ratio refers to a ratio of the number of verification results indicating verification approval to the number of voting validators.

8. The method according to claim 6, wherein, the verification consensus condition comprises: a voting verification score being greater than a verification score threshold corresponding to the business identifier;
wherein the voting verification score is determined based on verification results corresponding to all voting validators and voting weights, and the voting weights are determined based on credit scores of the validators.

9. The method according to any one of claims 6-8, wherein, a target threshold corresponding to the business identifier is determined based on at least one of: an economic value of the business operation, a risk level of the business operation, a user collateral asset ratio, a time urgency of source chain state changes, and customized business characteristics;
wherein the target threshold comprises a validator quantity threshold, a consensus ratio threshold, or a verification score threshold.

10. The method according to claim 9,wherein the validator quantity threshold, the consensus ratio threshold and the verification score threshold, are each:

- directly proportional to the economic value of the business operation;
- directly proportional to the risk level of the business operation;
- inversely proportional to the user collateral asset ratio; and
- inversely proportional to the time urgency of source chain state changes.

11. The method according to claim 2, wherein, after receiving the verification results returned by the validators, the method further comprises:
when verification results returned by validators within a preset verification time period do not satisfy the executable condition, executing an operation to expand the validator scope and transmitting

the verification request carrying the credential identifier to additional validators;
wherein the additional validators refer to newly selected validators participating in verification.

12. The method according to claim 2, wherein, after receiving the verification results returned by the validators, the method further comprises:
when verification results returned by validators within a preset verification time period do not satisfy the executable condition, executing a qualification upgrade operation and transmitting the verification request carrying the credential identifier to highly-qualified validators;
wherein the highly-qualified validators are validators whose current credit score is greater than a preset credit score or whose validator collateral asset ratio is greater than a preset collateral asset ratio.

13. The method according to claim 2, wherein, before transmitting the verification request carrying the credential identifier to validators, the method further comprises:
determining whether a verification request carrying the same credential identifier has been transmitted within a preset cooling-off period corresponding to the current time; if yes, stopping transmitting the verification request carrying the credential identifier to validators; if no, transmitting the verification request carrying the credential identifier to validators.

14. The method according to claim 2, wherein, after executing the business operation corresponding to the business identifier, the method further comprises:
updating a credential state mapping table and marking a current state corresponding to the credential identifier as a used state, wherein the used state is used to prevent the verifiable credential corresponding to the credential identifier from being reused.

15. The method according to claim 2, wherein, after receiving the verification results returned by the validators, the method further comprises:
determining a type corresponding to a voting validator based on an arrival time of the verification result;
when the type corresponding to the voting validator is a timely voting validator, providing a first incentive reward to the timely voting validator; wherein the timely voting validator is a validator

that returns a verification result before the executable condition is determined to be satisfied;
when the type corresponding to the voting validator is a late voting validator, providing a second incentive reward to the late voting validator or providing no incentive reward; wherein the late voting validator is a validator that returns a verification result after the executable condition is determined to be satisfied;
wherein the first incentive reward is greater than the second incentive reward.

16. The method according to claim 2, wherein, after receiving the verification results returned by the validators, the method further comprises:
determining a type corresponding to a voting validator based on the verification result and the verifiable data on the source chain;
when the type corresponding to the voting validator is an honest voting validator, increasing a credit score corresponding to the honest voting validator; wherein the honest voting validator is a voting validator whose verification result matches the verifiable data on the source chain;
when the type corresponding to the voting validator is a fraudulent voting validator, executing at least one of the following operations:
decreasing a credit score corresponding to the fraudulent voting validator, confiscating collateral assets corresponding to the fraudulent voting validator, and expelling the fraudulent voting validator from the validator network; wherein the fraudulent voting validator is a voting validator whose verification result does not match the verifiable data on the source chain.

17. A cross-chain state verification apparatus for blockchain, comprising:
an assertion obtaining module, configured to obtain a state assertion submitted by a user on a source chain, wherein the state assertion is an assertion regarding verifiable data on the source chain;
a credential generation module, configured to generate a verifiable credential, wherein the verifiable credential comprises a credential identifier, a decentralized identifier, the state assertion, and a content hash, wherein the content hash is a result of performing a hash operation on the state assertion;
a digital signature module, configured to digitally sign the verifiable credential based on a user private key corresponding to the decentralized identifier to obtain a digitally signed verifiable credential; and

a credential storage module, configured to store the digitally signed verifiable credential in a distributed storage system, such that a user on a destination chain can perform state verification on the verifiable data based on the digitally signed verifiable credential.

18. A cross-chain state verification apparatus for blockchain, comprising:
a request obtaining module, configured to obtain a cross-chain business request submitted by a user on a destination chain, wherein the cross-chain business request comprises a business identifier and a credential identifier;
a credential verification module, configured to transmit a verification request carrying the credential identifier to validators, such that the validators access a distributed storage system based on the credential identifier, determine a digitally signed verifiable credential, perform verification based on the verifiable credential, determine verification results, and receive the verification results returned by the validators;
an operation execution module, configured to execute a business operation corresponding to the business identifier when the verification results returned by the validators satisfy an executable condition;
wherein the verifiable credential comprises the credential identifier, a decentralized identifier, a state assertion, and a content hash; the content hash is a result of performing a hash operation on the state assertion; the digital signature is a result of digitally signing the content hash based on a user private key corresponding to the decentralized identifier; and the state assertion is an assertion regarding verifiable data on a source chain.

19. An electronic device comprising a memory, a processor, and a computer program stored in the memory and executable on the processor, wherein, when the processor executes the computer program, the cross-chain state verification method for blockchain according to any one of claims 1 to 16 is implemented.

20. A computer-readable storage medium storing a computer program, wherein, when the computer program is executed by a processor, the cross-chain state verification method for blockchain according to any one of claims 1 to 16 is implemented.